

MEMPHIS INTL, TN, USA

WMO: 723340

Lat: 35.056N

Lon: 89.987W

Elev: 254

StdP: 14.56

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13893

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.0	23.2	4.8	7.1	24.0	8.9	8.8	27.0	24.6	47.8	21.5	50.9	9.3	10	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	96.4	77.0	94.3	76.5	92.4	76.1	80.1	91.4	79.1	90.0	78.1	88.6	7.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.1	142.2	85.9	76.0	137.0	85.0	75.0	132.6	84.3	43.7	91.3	42.6	90.2	41.6	88.8	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.6	18.5	16.7	13.4	99.3	4.6	3.0	10.1	101.5	7.4	103.2	4.8	104.9	1.5	107.2
			11.4	81.9	4.2	1.4	8.3	82.9	5.9	83.7	3.5	84.5	0.4	85.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.5	42.0	46.1	54.2	63.5	72.2	79.9	82.4	82.2	76.2	64.8	53.0	44.9
	DBStd	16.64	11.27	10.72	10.05	8.09	6.70	4.67	3.94	4.34	6.34	8.45	9.10	10.00
	HDD50	863	300	186	71	7	0	0	0	0	0	3	73	223
	HDD65	2856	714	533	353	123	21	0	0	0	5	112	370	625
	CDD50	5801	52	75	202	410	688	897	1005	998	785	461	163	65
	CDD65	2321	2	3	19	76	244	447	540	533	340	105	10	2
	CDH74	24053	1	6	106	527	2015	4824	6287	6153	3254	828	48	4
	CDH80	10305	0	0	7	87	636	2136	2952	2932	1322	230	3	0
Wind	WSAvg	8.0	9.0	9.3	9.4	9.4	8.2	7.2	6.7	6.6	7.0	7.2	7.8	8.5
Precipitation	PrecAvg	51.3	3.7	4.2	5.4	5.2	5.2	3.5	3.8	3.4	3.6	3.1	5.2	5.5
	PrecMax	70.9	8.9	10.5	12.1	11.5	11.4	7.2	8.5	9.6	12.3	8.3	11.6	13.8
	PrecMin	34.4	0.6	1.1	1.5	1.6	0.8	0.5	1.2	0.6	0.6	0.0	0.7	1.2
	PrecStd	9.6	2.2	2.2	2.4	2.5	2.4	1.7	1.7	2.2	2.4	2.1	3.0	3.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.5	74.8	80.9	85.6	91.0	97.0	98.1	101.0	96.4	89.4	79.3	73.9
		MCWB	63.8	64.5	65.3	69.0	73.6	75.7	78.0	75.8	73.8	71.5	66.1	66.5
	2%	DB	67.6	70.8	77.1	82.6	88.4	94.3	95.7	96.9	93.0	85.5	74.5	68.5
		MCWB	61.9	62.1	63.2	67.3	72.3	75.7	77.7	77.4	74.5	70.1	63.5	62.3
	5%	DB	63.9	67.5	73.7	79.8	86.3	92.2	93.7	94.2	90.3	82.4	71.4	64.8
		MCWB	58.7	59.3	61.5	66.0	71.3	75.1	77.5	76.7	73.8	67.9	62.0	59.4
	10%	DB	59.9	63.6	70.0	76.9	83.8	90.0	91.7	91.8	87.5	78.8	67.8	61.0
		MCWB	54.0	56.2	59.4	64.8	70.6	74.7	77.0	76.1	72.1	66.4	60.1	56.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.6	67.3	68.1	72.8	77.2	80.2	81.3	81.6	78.2	75.5	69.3	67.7
		MCDB	70.0	71.7	76.4	81.4	86.3	90.6	92.7	93.5	89.5	84.9	74.3	72.0
	2%	WB	63.0	64.5	65.9	70.3	75.4	78.6	80.2	80.1	76.8	73.3	66.7	64.0
		MCDB	67.0	69.3	74.0	78.6	84.2	89.2	91.6	91.5	87.8	81.3	72.3	67.6
	5%	WB	59.8	61.4	63.6	68.7	73.8	77.6	79.3	78.8	75.6	71.2	64.3	60.5
		MCDB	63.6	66.3	71.1	76.7	82.5	88.3	90.5	89.6	85.7	78.4	69.6	64.1
	10%	WB	55.2	57.4	61.3	66.9	72.3	76.5	78.3	77.7	74.6	68.9	61.2	56.7
		MCDB	58.6	62.4	68.8	74.5	81.0	86.6	88.7	88.1	83.9	76.2	66.8	60.2
Mean Daily Temperature Range	5% DB	MDBR	16.0	16.8	18.1	18.6	17.5	17.2	16.4	17.2	18.4	19.1	17.8	15.5
		MCDBR	18.9	20.7	20.6	20.3	19.1	19.1	18.3	19.4	20.2	21.9	21.5	18.9
		MCWBR	15.0	14.6	12.0	10.1	8.1	6.8	6.3	6.2	7.1	10.0	13.7	14.6
	5% WB	MCDBR	17.1	18.3	17.8	16.7	16.8	16.8	16.8	16.9	16.9	17.6	18.0	16.5
	MCWBR	15.0	14.7	12.5	10.0	8.3	7.6	6.8	6.6	7.0	9.8	14.5	14.3	
Clear-Sky Solar Irradiance	taub	0.318	0.327	0.364	0.401	0.440	0.460	0.493	0.490	0.423	0.362	0.333	0.316	
	taud	2.496	2.467	2.392	2.284	2.234	2.187	2.130	2.121	2.304	2.451	2.483	2.517	
	Ebn at Noon	281	291	288	281	270	263	253	252	265	274	273	275	
	Edh at Noon	27	30	36	42	44	47	49	49	39	31	27	25	
All-Sky Solar Radiation	RadAvg	773	937	1302	1661	1867	2067	1980	1824	1557	1218	866	663	
	RadStd	65	126	117	109	113	123	100	116	113	147	89	67	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page